# FRANCESCO GANIS

## PhD, Sound & Music Engineer

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With a strong background in audio digital signal processing (DSP), prototyping, and the evaluation of audio solutions, I bring experience in developing user-centered audio technologies and vibrotactile feedback systems. My work has involved translating user and system needs into structured designs, defining experimental approaches, and iterating on prototypes based on data and user feedback. I am a passionate audio engineer and researcher, with deep love for music and sound. I am eager to contribute, while continuing to expand my knowledge and skills. You can find my portfolio at francescoganis.github.io.

## EXPERIENCE

### System Verification Engineer, Internship

**GN Hearing A/S**

Oct. 2026 – Ongoing | Ballerup, DK

Skills learned: User Research | Perceptual Testing | Technical Documentation

- Support the Early Adopters Program team with user-research projects
- Develop experimental paradigms for the evaluation of audio products
- Document and communicate findings through technical reporting and presentations for internal stakeholders

### Technology Evaluation Engineer, Internship

**Bang & Olufsen**

Apr. 2026 – May 2026 | Lyngby, DK

Skills learned: Product Evaluation | Perceptual Testing | Technical Documentation

- Development of experimental paradigm for the perceptual evaluation of high-end audio products
- Defined evaluation criteria, relevant attributes, and study procedures to support structured, evidence-based assessment
- Documented and communicated findings through technical reporting and presentations for internal stakeholders

### PhD, Tangible Interactive Technologies for Hearing and Music Development

**Aalborg University**

Sep. 2022 – Mar. 2026 | Copenhagen, DK

Skills learned: Research | Prototyping | Evaluating

- Designed and evaluated interactive audio and vibrotactile systems for children with hearing impairment
- Developed experimental approaches to assess system performance, user interaction, and training outcomes
- Translated user and clinical needs into system requirements, prototypes, and iterative design improvements
- Collaborated with audiologists and clinicians to align technical development with real-world use contexts
- Produced scientific and technical documentation of methods, system behavior, and results

## SKILLS

### Technical

**Audio**: DSP | MATLAB | MAX/MSP | Pure Data | Pro Tools | Reaper | Sound Engineering | Audio Testing

**Development**: C | Python | Git | Embedded Programming | Arduino | ESP32 | LaTeX | Unity | C#

**Prototyping & Fabrication**: Prototyping | Product Evaluation | Lasercutting | 3D Printing

### Professional

Analytical Thinking | Collaboration | Detail-oriented | Problem-solving | Scientific Dissemination | Supervision | Teamwork | Technical Documentation

## LANGUAGES

**English** ●●●●●
Full professional proficiency

**Italian** ●●●●●
Mother tongue

## PUBLICATIONS

### Thesis

- F. Ganis, "Tune in: Exploring music and sound training for children with hearing challenges," Ph.d.-serien for Det Tekniske Fakultet for IT og Design, Aalborg Universitet, 2025.

### Teaching Assistant, Sound & Music Computing

**Aalborg University**

Sep. 2022 – Aug. 2025 | Copenhagen, DK

Skills learned: Teaching | Communication | Supervision

- Supported teaching in the *Sound Processing* and *Physical Models for Sound Synthesis* courses
- Supervised student projects in the Medialogy and Sound & Music Computing programmes

### Student Helper, Sound Synthesis and Design

**Aalborg University**

Oct. 2021 – Apr. 2022 | Copenhagen, DK

Skills learned: MATLAB | Audio Programming | DSP

- Created sound examples in MATLAB for the book *Serafin, S., Buxton, B., Gaver, B., & Bly, S. (2022). Auditory interfaces. Focal Press*

### Intern, Prototype Development

**Oticon Medical**

Oct. 2021 – Jan. 2022 | Copenhagen, DK

Skills learned: Product Development | Prototyping | Evaluation

- Developed a vibrotactile smartphone prototype to support music listening for users with hearing impairment
- Translated user needs into functional prototype features in a hearing-technology setting
- Evaluated the system with attention to usability and perceptual experience

## EDUCATION

### Master's Degree, Sound & Music Computing

**Aalborg University**

Sep. 2020 – Jun. 2022 | Copenhagen, DK

Skills learned: DSP | Prototyping | Evaluating

- Thesis: "Vibrotactile Feedback for Musical Training of Cochlear Implants Users"
- Completed courses in Sound Processing, Embodied Interaction, Music Perception and Cognition, New Interfaces for Musical Expression

### Bachelor's Degree, Electronic Music & Sound Engineering

**"C. Pollini" Conservatory**

Sep. 2016 – Apr. 2020 | Padova, IT

Skills learned: Sound Engineering | MAX/MSP | Sound Design

- Graduated with honors
- Thesis: "Continuous Sonification of Autonomous Driving Vehicles with Purposes of Passengers Information and Trust Augmentation"

### Journal Articles

- F. Ganis et al., "The role of haptics in training and games for hearing-impaired individuals: A systematic review," *Multimodal Technologies and Interaction*, vol. 8, no. 1, p. 1, 2023.

### Conference Proceedings

- F. Ganis et al., "Vibrotactile memory: A case study of timbre perception training in children with cochlear implants using a video game," SMC Proceedings, 2024, pp. 134–141.
- F. Ganis et al., "Tickle tuner-haptic smartphone cover for cochlear implant users' musical training," in *HAID*, Springer Nature, 2022, p. 14.
- F. Ganis et al., "Real-time timbre transfer and sound synthesis using ddsp," SMC Proceedings, 2021, pp. 175–182.

## FURTHER TRAINING

### Masterclass, Live Electronics

**Chigiana Academy**

Jul. 2019 | Siena, IT

### Masterclass, Symphonic Orchestra Recording

**G. A. Fano Institute**

Jul. 2018 | Spilimbergo, IT

### Diploma, Music, Theory and Solfège

**G. Tartini Conservatory**

Sep. 2009 | Trieste, IT

## REFERENCES

**Oliver Scheuregger, Manager**
@ Bang & Olufsen, Lyngby - DK
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**Prof. Stefania Serafin, Supervisor**
@ Technical University of Denmark, Konges Lyngby - DK
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**Prof. Lone Percy-Smith, Supervisor**
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